INCH-POUND

FF-S-85F
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SUPERSEDING
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15-DEC-2022

# DETAIL SPECIFICATION SHEET

# SCREW, CAP, SLOTTED AND HEXAGON HEAD

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

Comments, suggestions, or questions on this document should be addressed to DLA Aviation VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or emailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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## 1. SCOPE AND CLASSIFICATION

1.1. Scope. This specification covers the types and styles of slotted and hexagon head cap screws, classified in 1.2.1 (see 6.1).

1.2. Classification.

1.2.1. Types and styles. Cap screws shall be of the following types and styles (see 6.2).

Type I - Slotted head.
Style 1s - Round head (see fig. 4)
Style 2s - Flat countersunk head, 82º (see fig. 4).
Style 4s - Fillister head (see fig. 4).

Type II - Plain head.
Style 10p - Hexagon head (see fig. 4).

1.2.2. Grades. Cap screws of steel material shall be of the following grades(see 6.2).

Grade 2 - Low carbon steel.
Grade 5 - Medium carbon steel.
Grade 8 - Alloy steel.
Grade 9 – High strength alloy steel.

1.2.3. Sizes. The sizes of cap screws shall be classified by the nominal body diameter from 1/4 inch thru 3 inches as specified (see 6.2).

## 2. APPLICABLE DOCUMENTS

2.1. General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2. Government Documents

2.2.1. Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-DTL-13924 – Coating, Oxide, Black, for Ferrous Metals



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MIL-DTL-16232 – Phosphate Coatings, Heavy, Manganese or Zinc Base

MIL-DTL-18240 – Fastener Element, Self-Locking, Threaded Fastener, 250˚F Maximum

MIL-F-495 – Finish, Chemical, Black, for Copper Alloys

MIL-PRF-8625 – Anodic Coatings for Aluminum and Aluminum Alloy

MIL-PRF-32647 – Zinc-Nickel Electroplating for Fasteners

MIL-STD-1916 – DoD Preferred Methods for Acceptance of Product

MS35295 – Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Medium Carbon Steel, Plain Finish UNC-2A

MS35302 – Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Alloy Steel, Plain Finish, UNF-2A

MS35309 – Screw, Cap, Hexagon Head - Naval Brass, Plain Finish, UNC-2A

MS35310 – Screw, Cap, Hexagon Head, Naval Brass, Plain Finish, UNF-2A

MS35311 – Screw, Cap, Hexagon Head, Nickel-Copper Alloy, Plain Finish UNC-2A

MS35312 – Screw, Cap, Hexagon Head, Nickel-Copper Alloy, Plain Finish UNF-2A

MS51099 – Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Head Drilled for Locking Wire, Steel, Corrosion-Resisting, Passivated, UNC-2A

MS51100 – Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Head Drilled for Locking Wire, Steel, Corrosion-Resisting, Passivated, UNF-2A

MS51107 – Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Shank Drilled for Cotter Pin, Alloy Steel, Grade 8, Phosphate Coated, UNC-2A

MS51108 – Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Shank Drilled for Cotter Pin, Alloy Steel, Grade 8, Phosphate Coated, UNF-2A

QQ-N-281 – Nickel-Copper Alloy Bar, Plate, Rod, Sheet, Strip, Wire, Forgings and Structural and Special Shaped Sections.

QQ-N-286 – Nickel-Copper-Aluminum Alloy, Wrought.

(Copies of the above documents are available online at https://quicksearch.dla.mil)



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2.3. Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1 – Unified Inch Screw Threads (UN and UNR Thread Form)

ASME B18.2.1 – Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)

ASME B46.1 – Surface Texture (Surface Roughness, Waviness and Lay)

(Copies of these documents are available for purchase online from https://www.asme.org/)

ASTM INTERNATIONAL

ASTM A342/A342M – Standard Test Methods for Permeability of Weakly Magnetic Materials

ASTM B21/B21M – Standard Specification for Naval Brass Rod, Bar, and Shapes

ASTM B138/B138M – Standard Specification for Manganese Bronze Rod, Bar, and Shapes

ASTM B139/B139M – Standard Specification for Phosphor Bronze Rod, Bar, and Shapes

ASTM B150/B150M – Standard Specification for Aluminum Bronze Rod, Bar, and Shapes

ASTM B154 – Standard Test Method for Mercurous Nitrate Test for Copper Alloys

ASTM B633 – Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel

ASTM D3951 – Standard Practice for Commercial Packaging

ASTM E10 – Standard Test Method for Brinell Hardness of Metallic Materials

ASTM E18 – Standard Test Methods for Rockwell Hardness of Metallic Materials

ASTM E112 – Standard Test Methods for Determining Average Grain Size



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ASTM E1282 – Standard Guide for Specifying the Chemical Compositions and Selecting Sampling Practices and Quantitative Analysis Methods for Metals, Ores, and Related Materials

ASTM E1417/E1417M – Standard Practice for Liquid Penetrant Testing

ASTM E1444/E1444M – Standard Practice for Magnetic Particle Testing for Aerospace

ASTM F468 – Standard Specification for Nonferrous Bolts, Hex Cap Screws, Socket Head Cap Screws, and Studs for General Use

ASTM F593 – Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs

ASTM F606/F606M – Standard Test Methods for Determining the Mechanical Properties of Externally and Internally Threaded Fasteners, Washers, Direct Tension Indicators, and Rivets

ASTM F788 – Standard Specification for Surface Discontinuities of Bolts, Screws, Studs, and Rivets, Inch and Metric Series

ASTM F2328 – Standard Test Method for Determining Decarburization and Carburization in Hardened and Tempered Threaded Steel Bolts, Screws, Studs, and Nuts

(Copies of these documents are available for purchase online from https://www.astm.org)

AMERICAN SOCIETY FOT QUALITY CONTROL (ASQ)

ASQ H1331 – Zero Acceptance Number Sampling Plans - Fifth Edition

(Copies of these documents are available for purchase online from https://asq.org).

AEROSPACE INDUSTRIES ASSOCIATION – NATIONAL AEROSPACE STANDARDS (AIA/NAS)

AIA/NAS NASM1312-6 – Fastener Test Methods - Method 6, Hardness

AIA/NAS NASM35307 – Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel Corrosion Resisting, Passivated, UNC-2A

AIA/NAS NASM35308 – Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel Corrosion Resisting, Passivated, UNF-2A

AIA/NAS NASM51095 – Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Head Drilled for Locking Wire, Steel, Grade 5, Cadmium Plated, UNC-2A, Plain and Self-locking



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AIA/NAS NASM51096 – Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Head Drilled for Locking Wire, Steel, Grade 5, Cadmium Plated, UNF-2A, Plain and Self-locking

AIA/NAS NASM51105 – Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Shank Drilled for Cotter Pin, Steel, Grade 5, Cadmium Plated, UNC-2A

AIA/NAS NASM51106 – Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Shank Drilled for Cotter Pin, Steel, Grade 5, Cadmium Plated, UNF-2A

AIA/NAS NASM51109 – Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Shank Drilled for Cotter Pin, Steel, Corrosion Resisting, Passivated, UNC-2A

AIA/NAS NASM51110 – Screw, Cap, Hexagon Head (Finished Hexagon Bolt, Shank Drilled for Cotter Pin, Steel, Corrosion Resisting Passivated, UNF-2A

AIA/NAS NASM90725 – Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNC-2A

AIA/NAS NASM90726 – Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNF-2A

(Copies of this document are available for purchase online from www.aia-aerospace.org)

SAE INTERNATIONAL

SAE-AIR4127 – Steel: Chemical Composition and Hardenability

SAE-AMS2461 – Plating, Zinc-Nickel Alloy (12 to 16% Ni)

SAE-AMS2700 – Steel, Passivation of Corrosion Resistant Steels

SAE-AMS4117 – Aluminum Alloy, Rolled or Cold Finished, Bars, Rods, and Wire and Flash Welded Rings 1.0Mg - 0.60Si - 0.28Cu - 0.20Cr (6061; -T6, -T651) - UNS A96061

SAE-AMS-QQ-A-225/6 – Aluminum Alloy, 2024, Bar, Rod, and Wire; Rolled, Drawn, or Cold Finished- UNS A92024

SAE-AMS-QQ-P-416 – Plating, Cadmium (Electrodeposited)

SAE-J429 – Mechanical and Material Requirements for Externally Threaded Fasteners

(Copies of these documents are available for purchase online from www.sae.org)



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2.4. Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. Material. Unless otherwise specified in the invitation for bids, contract or order, cap screws shall be made of materials specified in 3.1.1 through 3.1.5 and Table IA (see 6.2).

3.1.1. Low carbon steel. Unless otherwise specified (see 6.2) grade 2 cap screws shall be made of low carbon steel as shown in Table I. The chemical composition shall contain a maximum of 0.28 percent carbon, a maximum of 0.04 percent phosphorus, and a maximum of 0.05 percent sulfur.

3.1.2. Medium carbon steel. Grade 5 cap screws shall be made of medium carbon steel as shown in Table I and conforming to the requirements of SAE AIR4127. The chemical composition shall contain 0.28 to 0.55 percent carbon, a maximum of 0.04 percent phosphorus and a maximum of 0.05 percent sulfur. It shall be quenched and tempered at a minimum temperature of 800˚F. Cap screws of the following standards shall be made in accordance with the specification: MS35295, AIA/NAS NASM51095, AIA/NAS NASM51096, AIA/NAS NASM51105, AIA/NAS NASM51106, NASM90725, and AIA/NAS NASM90726.

3.1.3. Alloy steels.

3.1.3.1. Alloy steel, Grade 8. Grade 8 cap screws shall be made of alloy steel as shown in Table I and conforming to the requirements of SAE AIR4127. The chemical composition shall contain 0.28 to 0.55 percent carbon, a maximum of 0.04 percent phosphorus, and a maximum of 0.05 percent sulfur. Cap screws of the following standards shall be made in accordance with the specification: MS35302, MS51107, and MS51108.



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Table I. Grades and Mechanical Properties of Steel Cap Screws 0.25 threw 3 Inch Diameters

| Grade Designation | Nominal Size Dia., Inch | Proof Load p.s.i. | Tensile Strength p.s.i. (min) | Hardness | | | |
|---|---|---|---|---|---|---|---|
| | | | | Brinell No. | | Rockwell | |
| | | | | Min | Max | Min | Max |
| 2 | 0.25 thru 0.75* | 55,000 | 74,000 | 149 | 241 | B80 | B100 |
| | Over 0.75 thru 1.5 ** | 33,000 | 60,000 | 121 | 241 | B70 | B100 |
| 5 | 0.25 thru 1 | 85,000 | 120,000 | 255 | 321 | C25 | C34 |
| | Over 1 thru 1.5 | 74,000 | 105,000 | 223 | 285 | C19 | C30 |
| | Over 1.5 thru 3 | 55,000 | 90,000 | 183 | 235 | B90 | C21.7 |
| 8 | 0.25 thru 1.5 | 120,000 | 150,000 | 302 | 352 | C33 | C39 |
| | Over 1.5 thru 3 | 85,000 | 120,000 | 302 | 352 | C33 | C39 |
| 9 (High strength alloy) | 0.25 thru 1.5 | 130,000 | 170,000 | 336 | 386 | C36 | C45 |

* 6” and less in length.
** All diameters if length exceeds 6”.

3.1.3.2. Grade 9 alloy steel. The Grade 9 alloy steel shown in Table I shall be grade 8 conforming to the requirements of SAE AIR4127 and the chemical composition of paragraph 3.1.3.1. It shall meet the mechanical properties requirements specified in Table 1.

3.1.3.3. Alloy steels. Alloy steels shall be oil-quenched and tempered at a minimum temperature of 800˚F.

3.1.4. Corrosion resisting steels.

3.1.4.1. Austenitic. Austenitic corrosion resisting steel screws shall be manufactured from UNS S30000 series corrosion steel as specified in SAE AIR4127 and meet the mechanical properties specified in 3.2.2.1. Cap screws of AIA/NAS NASM35307 and AIA/NAS NASM35308 shall be made in accordance with the specification.

3.1.4.2. Ferritic and martensitic. Ferritic and martensitic corrosion resisting steel cap screws shall be manufactured from a steel composition such as steel numbers UNS S41000, UNS S41600, and UNS S43000 as specified in SAE AIR4127 and meet the mechanical properties specified in 3.2.2.2. Cap screws of MS51099, MS51100, AIA/NAS NASM51109, and AIA/NAS NASM51110 shall be made in accordance with the specification.

3.1.4.3. Galvanic corrosion. When corrosion resisting steels are used to fasten aluminum or carbon/alloy steel structures, the corrosion resisting steel fastener will cause accelerated corrosion of the structure due to galvanic effects; in this scenario it is recommended to coat corrosion resisting steel fasteners with zinc nickel per paragraph 3.4.8 in order to protect the aluminum or steel structures.

3.1.5. Non-ferrous material. Non-ferrous cap screws shall be manufactured from alloys specified in Table IA (see 6.2). Cap screws of MS35309, MS35310, MS35311, and MS35312 shall be made in accordance with the specification.



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3.1.5.1. Stress relief. Cold worked naval brass screws shall be stress relieved by heating the screws to a temperature of 600ºF and holding at that temperature for a minimum of one hour.

3.1.6. Self-locking element. Non-metallic, self-locking elements shall be in accordance with MIL-DTL-18240.

3.2. Mechanical properties.

3.2.1. Steel. Grades and mechanical properties of low carbon, medium carbon, alloy steel, and High strength alloy steel cap screws shall conform to Table I, as applicable (see 6.2).

3.2.2. Corrosion resisting steel.

3.2.2.1. Austenitic. Unless otherwise specified in the invitation for bids, contract or order, mechanical properties of austenitic corrosion-resisting steel cap screws shall conform to the following for diameters through 5/8 inch:

| | |
|---|---|
| a. Ultimate tensile strength | 80,000 p.s.i. min |
| b. Yield strength (0.2 percent offset) | 30,000 p.s.i. min. |
| c. Rockwell Hardness | B80 min. |
| d. Elongation | 30 percent min. |
| e. Reduction in area | 40 percent min. |

For diameters over 5/8 inch, ultimate tensile strength shall be 70,000 p.s.i. min. and have a hardness of Rockwell B74 min.

3.2.2.2. Ferritic and martensitic. Unless otherwise specified in the invitation for bids, contract or order, mechanical properties of ferritic and martensitic corrosion-resisting steel cap screws shall conform to the following:

| | |
|---|---|
| a. Ultimate tensile strength | 70,000 p.s.i. min |
| b. Yield strength (0.2% offset) | 40,000 p.s.i. min. |
| c. Brinell hardness | B235 max |
| d. Elongation | 15 |
| e. Reduction in area | 45 |

3.2.2.3. Nickel-Copper-Aluminum externally threaded fasteners. Nickel-Copper-Aluminum material that has been headed or threaded shall not be age hardened unless it has been solution annealed subsequent to the heading and threading operations. Threads formed after the final age hardening heat treatment shall be cut or ground only.

3.2.3. Non-ferrous cap screws. Unless otherwise specified in the invitation for bids, contract or order, mechanical properties of non-ferrous cap screws shall conform to Table IA, as specified (see 6.2).



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Table IA. Mechanical Properties of Screws of Non-Ferrous Materials

| Material | Applicable Specifications | UNS No. or class | Temper code or condition | Tensile strength p.s.i minimum | Yield strength at 0.2 percent offset or at extension indicated | | Elongation minimum |
|---|---|---|---|---|---|---|---|
| | | | | | Minimum p.s.i | Extension under load | |
| Manganese bronze | ASTM B138/B138M | C67500 | O60 | 55,000 | 22,000 | 0.5 percent | 20 |
| Aluminum bronze | ASTM B150/150M | C36200 | TQ55<br>O20<br>O25 | 90,000 | 37,000 | 0.5 percent | 18 |
| Phosphor bronze | ASTM B139/139M | C51000 | H04 | 60,000 | 33,000 | --- | 15 |
| Silicon bronze | ASTM F468 | C65100 | --- | 60,000 | 40,000 | 0.5 percent | 10 |
| Naval brass | ASTM B21/B21M | C64200 or C46400 | H04<br>Or<br>O50 | 60,000 | 27,000 | 0.5 percent | 25 2/ |
| Aluminum alloy 2024 | SAE-AMS-QQ-A-225/6 | --- | Solution heat-treated Temper T4 | 62,000 | 40,000 | 0.0058 inch per inch | 10 3/ |
| Aluminum alloy 6061 | SAE AMS4117 | --- | Age hardened temper T6 | 42,000 | 35,000 | 0.0055 inch per inch | 10 3/ |
| Nickel-copper alloy | QQ-N-281 | Class A | | 80,000 | 40,000 | 0.0071 inch in 2 inches | 20 1/ |
| Nickel-copper aluminum alloy | QQ-N-286 | Class A | Form 1 | 130,000 | 90,000 | 0.0109 inch in 2 inches | 20 1/ |

1/ In 2 inches (min)
2/ In 4 times diameter or thickness (min)
3/ In 2 inches or 4 times diameter (min)

3.3. Dimensions and tolerances. Cap screws shall conform to the dimensions and tolerances of ASME B18.2.1 except as specified herein.

3.3.1. Lengths. Unless otherwise specified (see 6.2), cap screws shall be furnished in the following length increments:

Lengths up to 0.5 inch incl – 0.0625 inch increments.
over 0.5 to 1 inches incl – 0.125 inch increments.
over 1 to 5 inches incl – 0.25 inch increments.
over 5 inches – 0.5 inch increments.

3.3.2. Thread series, class, lengths, and hand. Unless otherwise specified (see 6.2), cap screws shall be right - hand of the UNC (coarse thread) or UNF (fine thread) series, Class 2A, in accordance with ASME B1.1.



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3.3.3. Threads protectively coated. Unless otherwise specified (see 6.2), threads protectively coated shall meet the dimensions of ASME B1.1 after the coating has been applied.

3.3.4. Bearing surface. Bearing surface of cap screw heads, except flat head, shall be at right angles to the axis of the body within a tolerance of 2 degrees and of hex head within a tolerance of 2 degrees for 1 inch size and smaller, and 1 degree for sizes larger than 1 inch. The axis of the head shall be concentric with the axis of the body (determined by one diameter length of body under head) within a tolerance equal to 3 percent (6 percent TIR) of maximum width across flats. Any mutilation of the bearing surface sufficient to prevent proper seating of the screw shall be cause for rejection.

3.4. Protective coatings.

3.4.1. Coatings. Cap screws shall be furnished uncoated, passivated, or coated as specified herein (see 6.2).

3.4.2. Passivation. Corrosion resistant steel cap screws shall be passivated in accordance with SAE-AMS2700, Method 1.

3.4.3. Cadmium plating. When specified (see 6.2), cap screws shall be cadmium plated in accordance with SAE-AMS-QQ-P-416, Type II, Class 2. Cadmium is an R1, R2, and D1 substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per paragraph 3.4.8 is the preferred alternate.

3.4.4. Zinc plating. When specified (see 6.2), cap screws shall be zinc plated as specified in ASTM B633, type II, class FE/ZN 5. Zinc plating is not recommended for high strength steels or for aerospace applications.

3.4.5. Anodizing. Unless otherwise specified, aluminum alloy cap screws shall be electrolytically treated in conformance to MIL-PRF-8625, type II, class 1, non- dyed (see 6.2).

3.4.6. Phosphate. When specified (see 6.2), cap screws shall be treated in conformance to MIL-DTL-16232, type Z, class 2.

3.4.7. Black chemical. When specified (see 6.2), brass cap screws shall be treated with a black chemical finish in accordance with MIL-F-495.

3.4.8. Zinc Nickel plating. When specified (see 6.2), cap screws shall be zinc nickel plated in accordance with SAE-AMS2461, Type II, Class 2 or MIL-PRF-32647, minimum thickness 0.0003 inches. Zinc Nickel plating was developed as an environmentally friendly alternative to cadmium plating and is recommended for use in lieu of cadmium plating.

3.4.9. Black oxide coating. When specified (see 6.2), steel cap screws shall be treated with a black chemical finish in accordance with MIL-DTL-13924, Class 1.



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3.4.10. Hydrogen embrittlement. Alloy steel cap screws required to be electroplated or phosphate coated (see 3.4.3 thru 3.4.6) shall be subjected to a relief treatment in accordance with the methods specified in the applicable plating or coating specification, immediately after the plating or coating operation to minimize the embrittlement that results from these processes.

3.5. Metallographic requirements.

3.5.1. Decarburization. When specified (see 6.2), results shall be in accordance with the requirements of ASTM F2328.

3.5.2. Grain size of copper alloy. When specified (see 6.2), grain size of copper alloy in cap screws shall be an option of the contracting officer in accordance with the material specification.

3.6. Surface roughness. Unless otherwise specified (see 6.2), cap screws shall have a maximum arithmetical average surface roughness on the bearing surface of the head and shank of 125 microinches. Hot headed cap screws over 3/4 inch in diameter are excepted and shall have a maximum surface roughness of 250 microinches. Threads shall have a maximum surface roughness of 63 microinches. Normally, it will be sufficient to ascertain that these surfaces on screws have the equivalent of a smooth machined finish by visual comparison with known surface standards. However, then it is deemed necessary to measure these surfaces with commercially available equipment, roughness measurements shall be taken axially on the body and fillet surfaces, and circumferentially on the bearing surface in accordance with ASME B46.1.

3.7. Method of manufacture. Unless the method of manufacture is specifically stated (see 6.2), the method employed for the production of screw threads on cap screws shall be at the option of the manufacturer.

3.8. Head markings. Steel cap screws shall be permanently marked with the grade identification symbol conforming to SAE J429, and applicable documents. In addition, all cap screws shall be marked with the manufacturer's identification symbol. Markings shall be located on the top of the head, and may be either raised or depressed, at the option of the manufacturer. Grade 9 steel cap screws shall be permanently marked with the grade identification symbol FFS85_9.

3.9. Defects and Discontinuities. Defects and discontinuities shall meet the requirements of ASTM F788.

4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. Inspection records of the examination and tests with itemized results shall be kept complete at the manufacturer's facility, available to the Government throughout the duration of the contract, or a minimum of two years, whichever is longer. The Government reserves the right to perform any of the inspections set forth in the specification



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where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1. Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing requirements, is an acceptable practice to certain conformance to requirements, however, this does not authorize submission of known defective material either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2. Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in application referenced documents.

4.1.3. Examination of preparation for delivery. An examination shall be made to determine that the packaging, packing, and marking comply with the requirements in Section 5 of this specification.

4.2. Sampling for lot acceptance.

4.2.1. Inspection lot. A lot shall consist of cap screws of the same type, made from the same heat of material, heat treatment, protective coating, surface finish, size, length, and threads produced and manufactured at the same time by the same manufacturer's facility and offered for acceptance at the same time. Where the length of the lot of fasteners is too short to support mechanical testing of full size fasteners, the lot shall include longer length fasteners solely for the purposes of mechanical testing. The longer test fasteners shall be made from the same heat of rod, bar or wire used for the production fasteners; have the same type, style and diameter as the production fasteners; and shall be processed and heat treated with the production fasteners.

4.2.2. Sampling for examination of cap screws. Unless otherwise specified in the contract (see 6.2), for sampling for examination of cap screws, an "accept at zero" (c=0) sampling plan such as MIL-STD-1916 or ASQ H1331 shall apply. A random sample of cap screws shall be selected from each lot in accordance with MIL-STD-1916 or ASQ H1331 using the applicable Verification Level or Index Value listed in Table II.



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Table II. Classification of Defects

| Categories | Defects | Inspection Method |
|---|---|---|
| Critical<br>1 | Verification Level IV, Index Value 1.0<br>Cracks (3.9.3) | <br>Microscope or S.I.E.* |
| Major<br>101<br><br>102<br>103<br>104 | Verification Level III, Index value = 2.5<br>Thread size and form (3.3.1 and 3.3.2)<br><br>Thread lengths (3.3.2)<br>Discontinuities (3.9.1)<br>Self-locking element missing, when applicable (3.1.6) | <br>Thread ring gauge Go and No Go<br>S.I.E.*<br>S.I.E.*<br>Visual |
| Minor<br><br>201<br>202<br>203<br>204 | Verification Level II, Index value = 4.0 percent<br>Length (3.1.3)<br>Surface roughness (3.6)<br>Coated threads (when applicable)<br>Head marking (3.8) | <br><br>S.I.E.*<br>Visual<br>Visual<br>Visual |

*Standard Inspection Equipment

4.2.3. Sampling for mechanical properties and metallographic tests. Unless otherwise specified in the contract (see 6.2), for sampling for mechanical properties and metallographic tests, an "accept at zero" (c=0) sampling plan such as MIL-STD-1916 or ASQ H1331 shall apply. A random sample of cap screws shall be selected from each lot in accordance with MIL-STD-1916, verification level IV or ASQ H1331 index value 1.5.

4.2.3.1. Samples selected as specified in 4.2.3 may be used when practicable. Three quarter inch and larger test specimens processed with the lot may be used for destructive testing.

4.2.4. Chemical analysis. Chemical analysis shall be made in accordance with ASTM E1282. When a certificate covering these requirements can be furnished, it will be acceptable.

4.2.5. Sampling for packaging and packing. Sampling for inspection of packaging and packing shall be in accordance with ASTM D3951.

4.2.6. Sampling for mercurous nitrate test. Unless otherwise specified in the contract (see 6.2), for sampling for mercurous nitrate testing, an "accept at zero" (c=0) sampling plan such as MIL-STD-1916 or ASQ H1331 shall apply. A random sample of cap screws shall be selected from each lot in accordance with MIL-STD-1916 verification level VII or ASQ H1331 index value 0.1.



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4.3. Examination.

4.3.1. Visual and dimensional inspection. Each cap screw selected as specified in 4.2.2 shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with table II. Any screw having one or more defects shall be rejected and if the number of defective screws in any sample exceed the acceptance number of the sample, the lot represented by the sample shall be rejected. Retests shall be in accordance with ASQ Z1.4.

4.3.2. Inspection of packaging and packing. Inspection of packaging and packing including preservation and marking of packages and containers shall be in accordance with ASTM D3951.

4.4. Test procedure.

4.4.1. The test specified in 4.4.2 through 4.4.13 are all required. If wedge testing Is desired, see Appendix 1.

4.4.2. Hardness. Hardness test for cap screw shall be conducted in accordance with ASTM E10 and ASTM E18 on transverse section through the threaded portion of the screw, taken one nominal diameter from the point, or when too short, halfway between the bearing surface and the point. Hardness shall be measured at a point on the section, one quarter diameter from the axis of the screw. For screws three quarter inch and larger, test may be made on a test specimen.

4.4.3. Tensile. Tensile test shall be conducted in accordance with ASTM F606/F606M. Unless otherwise specified (see 6.2), Grade 2, low carbon steel screws shall not be subjected to this test.

4.4.3.1. Full size specimens. When testing equipment is available cap screws shall be tested full size, preferably using apparatus similar to that illustrated in figure 1.

4.4.3.2. Turned shank specimens. For testing cap screws having tensile properties greater than can be measured on available test equipment, the shank or test specimen shall be turned down around the axis, to gage dimensions specified in ASTM F606/F606M. For testing cap screws or test specimens too large for full size testing, the specimen shall be turned, have its axis midway between the axis of the shank and the surface of the shank, as illustrated in figure 3.

4.4.3.3. Ductility test specimens. Test for elongation and reduction in area shall be conducted on turned shank specimens (figure 2) having dimensions to the largest practical specimens (see 4.4.3).

4.4.3.4. Screws too short to be tensile strength tested as defined in ASTM F606/F606M shall be hardness tested in accordance with AIA/NAS NASM1312-6. If the hardness limits are not specified, screws shall meet the minimum hardness requirement of the material from which they were made.

4.4.3.5. Sampling for fasteners too long for testing. For lots of fasteners too long for testing in accordance with ASTM F606/F606M, standard length fasteners manufactured with the same lot



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to support mechanical testing shall be utilized to support proof, yield, axial and wedge strength testing, but shall not be used for evaluation of overall screw length, thread length or grip length.

Metallographic tests.

4.4.3.6. Decarburization. When required (see 6.2), tests for decarburization shall be performed in accordance with ASTM F2328.

4.4.3.7. Grain size of copper alloys. Determination of grain size of copper alloy shall be made in accordance with ASTM E112.

4.4.4. Protective coating test. Protective coating and plating tests shall be conducted in accordance with the applicable specifications shown in 3.4.3 through 3.4.7.

4.4.5. Passivation test. Passivation test shall be conducted in accordance with SAE-AMS2700, Method 1.

4.4.6. Hydrogen embrittlement. The contractor shall furnish the Government certification that electroplated, or phosphate coated alloy steel screws have been subjected to the hydrogen embrittlement relief treatment specified in 3.4.10. When specified (see 6.2), electroplated or phosphate coated screws shall be subjected to the embrittlement relief test specified in the applicable plating or coating specification.

4.4.7. Magnetic permeability. Austenitic corrosion resisting steel cap screws, subjected to visual and dimensional examination, shall also be tested to determine magnetic permeability (see 3.1.4.1) in accordance with ASTM A342/A342M.

4.4.8. Discontinuities. Magnetic particle inspection performed in accordance with ASTM E1444/E1444M or liquid penetrant inspection performed in accordance with ASTM E1417/E1417M shall be used to determine the presence of cracks and discontinuities such as laps, seams, and inclusions. Magnetic particle or penetrant indications alone shall not be cause for rejection, and such screws may be sectioned, and discontinuities measured microscopically under 10X magnification to determine conformance to the requirements of 3.9. The inspection shall be performed on unfinished cap screws, free of lubrication and subsequent to any processing operation which could adversely affect the cap screws. The acceptance of these discontinuities shall be as defined in ASTM F788.

4.4.9. Self-locking. Self-locking screws having non-metallic elements shall be tested in accordance with MIL-DTL-18240.

4.4.10. Mercurous nitrate (stress relief) test. The contractor shall furnish the government certification that cold worked naval brass screws have been subjected to the stress relief treatment specified in 3.1.5.1. When specified (see 6.2), samples taken as specified in 4.2.6 shall be subjected to a mercurous nitrate test conducted in accordance with ASTM B154/B154M. There shall be no evidence of cracks.



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4.4.11. Proof load. Proof load test for cap screws shall be conducted in accordance with ASTM F606/F606M.

4.4.12. Yield strength. Yield strength test for cap screws shall be conducted in accordance with ASTM F606/F606M.

4.4.13. Wedge tensile tests. Wedge tensile tests are only required by manufacturers when required by contract. Wedge tensile strength test for steel cap screws, when required, shall be conducted in accordance with ASTM F606/F606M. Wedge tensile strength test for corrosion resisting steel cap screws, when required, shall be conducted in accordance with ASTM F606/F606M using the wedge angles of ASTM F593. Wedge tensile strength test for non-ferrous cap screws, when required, shall be conducted in accordance with ASTM F606/F606M using the wedge angles of ASTM F468. Flat countersunk head screws cannot be wedge tensile tested, even when required.

## 5. PACKAGING

5.1. Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

6.1. Intended use. Hexagon head and slotted head cap screws are designed primarily for use in precision machine tools and similar applications. They are commonly used in tapped holes but may be used with nuts.

6.2. Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and style, grade (for carbon and alloy steel), and size or standard part number (1.2).
(c) Material and condition (3.1).
(d) Mechanical properties, if other than those specified in Tables I and IA (3.2).
(e) Length, thread series, class, and hand (3.3).
(f) Coating, if required, and whether thread shall conform to different size requirements (3.3.3 and 3.4).
(g) Decarburization test, if required (3.5.1).
(h) Copper alloy grain size, if required (3.5.2).
(i) Surface roughness, if different (3.6).
(j) Method of manufacture, if necessary (3.7).



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(k) Material certification, if applicable (4.2.4).
(l) Hardness and tensile tests (4.4.2 and 4.4.3).
(m) Metallographic tests, if required (4.4.4).
(n) Selection of applicable level of packaging and packing required (5.1).
(o) Embrittlement relief test, if required (4.4.7).
(p) Sampling for examination of cap screws (4.2.2).
(q) Sampling for mechanical properties and metallographic tests (4.2.3).
(r) Mercurous nitrate test, if required (4.4.10).
(s) Sampling method for Mercurous nitrate test, if required (4.2.6).
(t) Proof load test, if required (4.4.11).
(u) Yield strength test, if required (4.4.12).
(v) Wedge test, if required (4.4.13).

6.3. Military procurement. Items procured under this specification for Military use shall be limited to the variety and materials specified on the applicable MS. Cap screws of low carbon steel are non-preferred for military use. Personnel of the Military departments are requested to refer to these documents for guidance.

6.4. Definitions.

6.4.1. Cracks. A crack is a clean crystalline break passing through the grain or grain boundary without the inclusion of foreign elements.

6.4.1.1. Quench cracks. Quench cracks are fractures occurring during heat treatment, due to excessive high thermal and transformation stresses.

6.4.1.2. Forging cracks. Forging cracks are cracks that may occur during the cut-off or forging operations.

6.4.2. Lap. A lap is a surface defect appearing as a seam, caused by the folding over of metal fins or sharp corners and then rolling or forging them into the surface, but not welding them.

6.4.3. Seam. A seam is an unwelded fold or lap which appears as an opening in the raw material as received from the source.

6.4.4. Inclusions. Inclusions are non-metallic material in a solid metallic matrix.

6.4.5. Tool marks. Tool marks are longitudinal or circumferential grooves of shallow depth produced by the movement of manufacturing tools over the surface of the screw.

6.4.6. Void. A void is a shallow pocket or hollow on the surface of the screw due to non-filling of metal during forging or upsetting.

6.4.7. Fold. A fold is a doubling over of metal which may occur during the forging operation.

6.4.8. Burst. A burst is an open break in the metal.



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6.5. Subject term (keyword) listing.

Slotted head screw
Plain head screw
Screw thread

6.6. Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.



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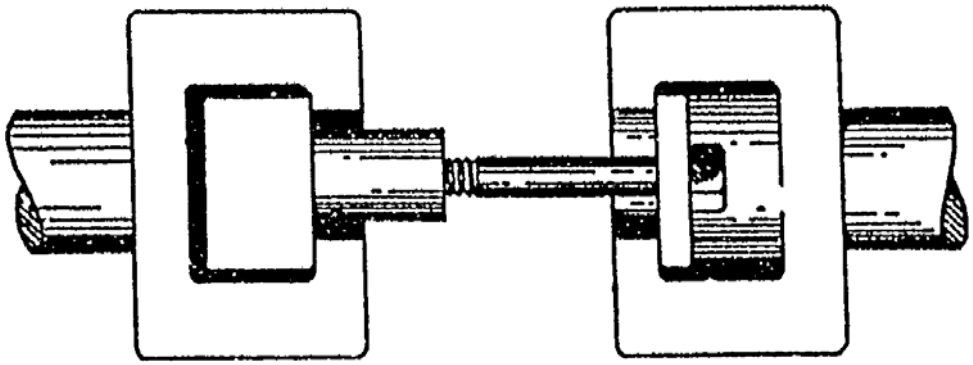


Figure 1. Tension Testing Full-Size Screw

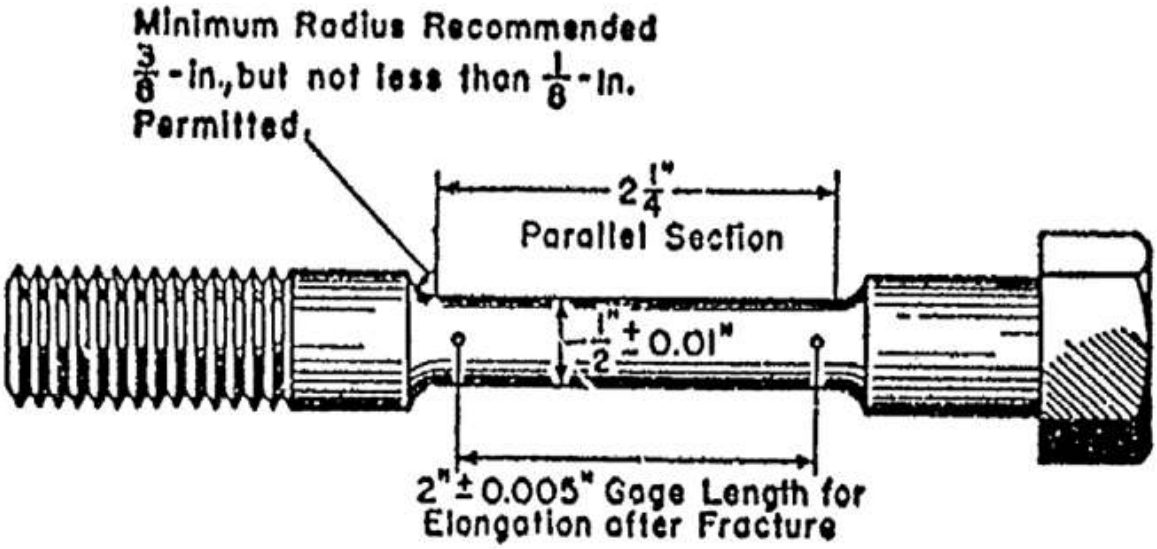



Figure 2. Tension-Test Specimen for Screw with Turned-Down Shank

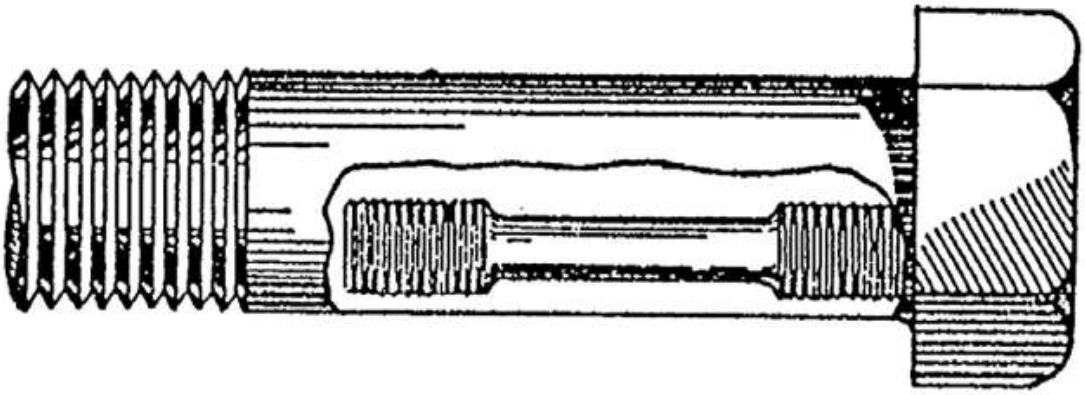


Figure 3. Location of Standard, Round, 2-inch-gage-length, Tension-Test Specimen when Turned from Large-Size Screw



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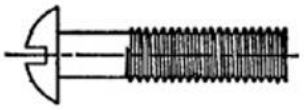
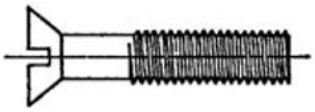
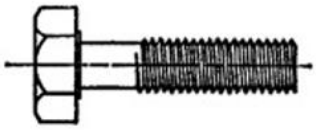
| TYPE | STYLE | HEAD | DRIVE |
|---|---|---|---|
| I | 1s | ROUND |  |
| I | 2s | FLAT |  |
| I | 4s | FILLISTER |  |
| II | 10P | HEXAGON |  |

Figure 4. Types, Heads and Styles



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Custodians:
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Navy – SH
Air Force – 71
DLA – GS

Review Interests:
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Other DoD – NS

Preparing Activity:
DLA – GS7

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